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Part 1: Label and Analyze the Data

Part 2: Train a Model

Reading: Your Tasks for Part 2

Graded Assignment: Training a Model

Reading: Our trained model

Part 3: Evaluate Your Model

Summary and Next Steps

Assignment details

Due

Mar 10, 11:59 PM IST

Attempts

Unlimited

Submitted

Mar 1, 1:30 AM IST

Your grade

To pass you need at least 100%. We keep your highest score.

100%

Instructions

1. There are 400 images in the "Train" folder for the parking signs dataset. You need to split this into training and validation sets. How many images did you use in your training set?

1 / 1 point

Try again

320

Correct

There is no correct split for training and validation data though somewhere between 70-90% is often used.

View submission

See feedback

2. Assume you used a mini-batch size of 48 and 300 training images for this data. What value below is an appropriate validation frequency? Recall that it is good practice to perform validation one or two times per epoch.

1 / 1 point

1

6

20

40

Correct

An epoch is six iterations using the above settings, so this value will calculate the validation loss every epoch.

3. When creating your detector, you should set the InputSize option to be the size of the image you're using for training. What size did you set for this option?

1 / 1 point

1 detector = yoloxObjectDetector("tiny-coco",classes,"InputSize=ImageSize");

[320 480]

[480 320]

[320 480 3]

[480 320 3]

Correct

This is the size of the images.

4. This image is the loss curve we had when training a detector using the recommended settings in the previous reading: mini-batch size = 16, maximum epochs = 10. Looking at the curve, what changes would you recommend?

1 / 1 point

140

120

100

80

60

40

20

0

0

20

40

60

80

100

120

140

160

180

Information

Epoch: 10

Duration: 100

Loss: 0.001

Correct

Yes. The loss is still decreasing when training ends, so increasing the maximum epochs is a good idea. Also, the iteration-to-iteration variation is a bit large. A larger mini-batch size may help.

5. Were you able to successfully train and test your own model for this project (all answers are correct)?

1 / 1 point

Yes, I trained a model and tried tested on several images and am happy with the results.

Yes, I trained a model, but it doesn't seem to work as well as I like.

No, I received errors and was not able to successfully train a model.

No, I took this quiz to get the pre-trained model and complete the course.

Feedback

Good job successfully training a model. Move on to evaluation to quantify how well your model performs. Remember that training is an iterative process. You'll often need to train multiple models and try various approaches to good results.